



United International University
Course: CSE 4783 - Cryptography, Spring 2026
Class Test 4, Set A
Total Marks: 20

Name		ID		Section	A
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1. In a (3,5)-threshold Shamir Secret Sharing scheme over $GF(17)$, the dealer distributes the following shares among five participants: (1,13), (2,6), (3,3), (4,4), and (5,9). Using any three of these shares and applying Lagrange interpolation, reconstruct the secret $S=f(0)$. Show all intermediate calculations. [5]

2. What is Zero Knowledge Proof (ZKP)? Mention the three properties of ZKP. [5]

3. Explain the Diffie-Hellman Key Exchange with Certificates using a diagram. [5]

4. Explain the Schnorr Identification Protocol with a diagram. [5]